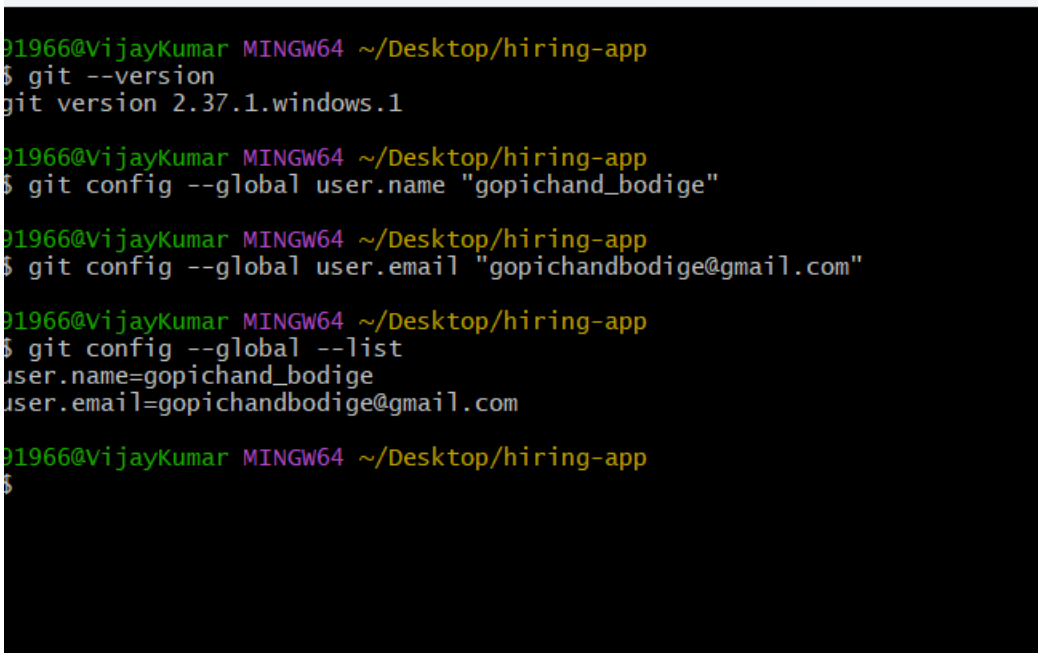
4. Git configuration

git config --global user.name

This command sets the username that Git attaches to every commit. It is used so that Git knows who made each change. The --global option saves it once for all repositories of the current user.

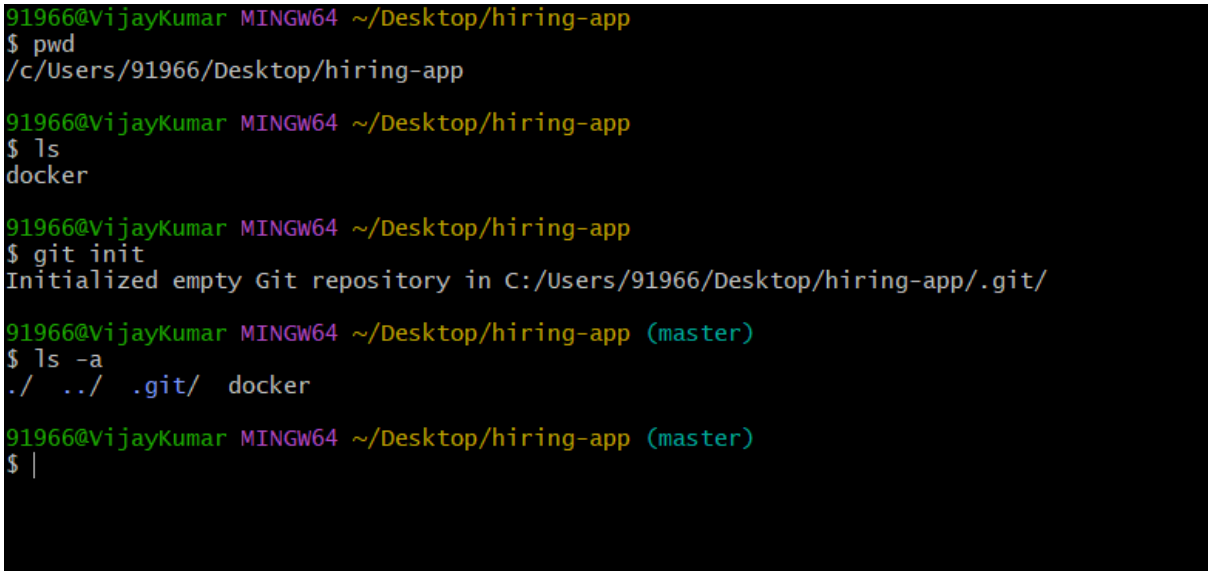
git config --global user.email

This command sets the email ID that Git attaches to every commit. It is used to identify the author of the changes, along with the username. The --global option saves it once for all repositories of the current user.



--->Git init

git init is a Git command that turns a normal folder into a Git repository by creating a hidden .git folder inside it. This folder stores all the version history and settings of the project.



-→ git status

git status is a Git command that shows the current state of the files in a repository. It tells which files are new (untracked), which files are modified, and which files are added to the staging area and ready to be committed.

→touch a

touch a is a Linux command that creates a new empty file named a in the current directory. If a file with the name a already exists, the command does not delete or change its content, it only updates the file's last modified time.

```
91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ ls -a
./  ../  .git/  docker

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ cd .git

91966@VijayKumar MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$ ll
total 7
-rw-r--r-- 1 91966 197609  23 Oct 22 16:24 HEAD
-rw-r--r-- 1 91966 197609 130 Oct 22 16:24 config
-rw-r--r-- 1 91966 197609  73 Oct 22 16:24 description
drwxr-xr-x 1 91966 197609   0 Oct 22 16:24 hooks/
drwxr-xr-x 1 91966 197609   0 Oct 22 16:24 info/
drwxr-xr-x 1 91966 197609   0 Oct 22 16:24 objects/
drwxr-xr-x 1 91966 197609   0 Oct 22 16:24 refs/

91966@VijayKumar MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$ cd..
bash: cd..: command not found

91966@VijayKumar MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$ cd ..

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        docker

nothing added to commit but untracked files present (use "git add" to track)

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ touch a

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        docker

nothing added to commit but untracked files present (use "git add" to track)

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
```

→ **git add .**

git add . is a Git command that adds all the new and modified files of the current directory (and its subdirectories) to the staging area. The dot (**.**) means the current directory.

```
91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git add .

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   docker


91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$
```

→ **Creating multiple files (b, c, d, e, f, g, h, i, j)**

We can create many files at once using the touch command by giving all the file names in a single line, separated by spaces. Each name creates a new empty file in the current directory.

```
91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ touch b c d e f g h i j k l m

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ ls
a  b  c  d  docker  e  f  g  h  i  j  k  l  m

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   docker

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        b
        c
        d
        e
        f
        g
        h
        i
        j
        k
        l
        m


91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git add .

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$
```

```
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   b
        new file:   c
        new file:   d
        new file:   docker
        new file:   e
        new file:   f
        new file:   g
        new file:   h
        new file:   i
        new file:   j
        new file:   k
        new file:   l
        new file:   m


91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$
```

→ git commit -m “ ”

git commit -m is a Git command that saves the staged changes into the repository as a new version (commit). The -m option lets us write a short message inside the quotes that describes what was changed, for example “Added files b to j”.

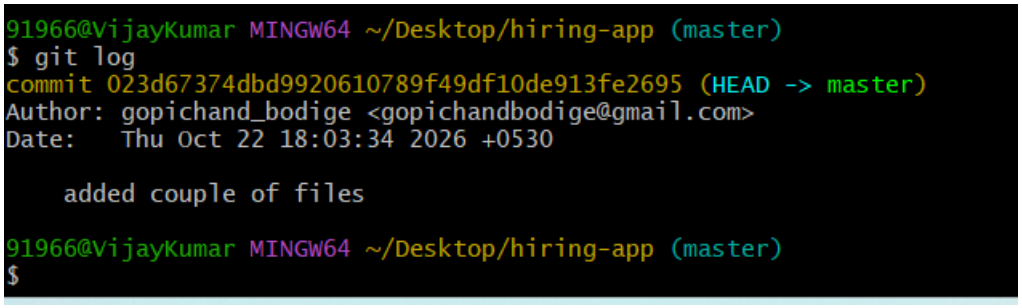
```
91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "added couple of files"
[master (root-commit) 023d673] added couple of files
 14 files changed, 7 insertions(+)
 create mode 100644 a
 create mode 100644 b
 create mode 100644 c
 create mode 100644 d
 create mode 100644 docker
 create mode 100644 e
 create mode 100644 f
 create mode 100644 g
 create mode 100644 h
 create mode 100644 i
 create mode 100644 j
 create mode 100644 k
 create mode 100644 l
 create mode 100644 m

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master
nothing to commit, working tree clean

91966@VijayKumar MINGW64 ~/Desktop/hiring-app (master)
$
```

→ git log

git log is a Git command that shows the history of all the commits made in the repository, starting from the latest one. For each commit, it displays the commit ID, the author's name and email, the date and time, and the commit message.

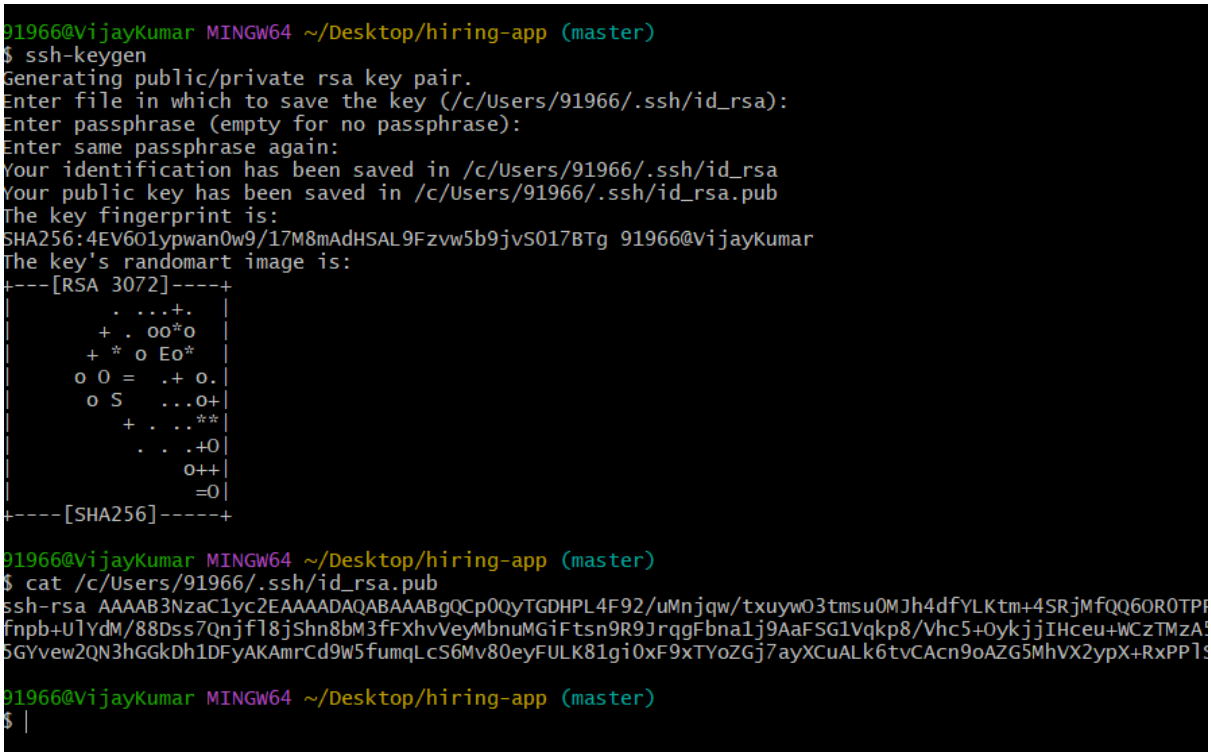


→ssh-keygen

ssh-keygen is a Linux command that generates an SSH key pair, which consists of a private key and a public key. These keys are used to log in to servers or to connect to services like GitHub securely, without typing a password every time.

It is used for secure authentication. The private key stays on our machine and must never be shared, and the public key is copied to the server or added to GitHub. When we connect, the server checks that the private key matches the public key and allows access.

- **id_rsa** (or id_ed25519): the private key, which must be kept secret.
- **id_rsa.pub** (or id_ed25519.pub): the public key, which can be shared.



SSH keys

New

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

laps
SHA256:4EV6O1ypwan0w9/17M8mAdHSAL9Fzvw5b9jvS017BTg
SSH
Added on Oct 22, 2026
Never used — Read/write

Check out our guide to connecting to GitHub using SSH keys or troubleshoot common SSH problems.

GPG keys

New

There are no GPG keys associated with your account.

Learn how to generate a GPG key and add it to your account.

Vigilant mode

☐ **Flag unsigned commits as unverified**
This will include any commit attributed to your account but not signed with your GPG or S/MIME key.

→git push

git push is a Git command that uploads the commits from the local repository (our server) to a remote repository, such as GitHub. Only the committed changes are sent, so files that are not committed are not uploaded.

